

MOHAMED NAGY

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OBJECTIVE

A passionate Software Engineer with a strong foundation in open-source technologies, Linux system programming, and full-stack development. Experienced in building scalable applications using modern frameworks, contributing to open-source projects, and applying software engineering best practices including CI/CD, version control, and Agile methodologies. Seeking a full-time position to leverage my skills in designing, developing, and maintaining high-quality software solutions in a collaborative, open-source-driven environment.

EDUCATION

ITI 9-Months Program

2025 – 2026

Open Source Application Development

Smart Village, Egypt

Alexandria University

2020 – 2025

B.Sc. in Electronics and Communication Engineering

CGPA: 3.36 (Excellent)

- **Graduation Project:** *Advanced Driver Assistance Systems (ADAS)*
- Designed and implemented an **Adaptive Front Lighting System (AFS)** that dynamically adjusts LED headlights based on real-time lane detection to enhance nighttime driving safety.
- Developed and maintained the CI/CD pipeline for the **FOTA (Firmware Over-The-Air)** system, automating firmware delivery and validation processes for automotive ECUs, ensuring efficient and reliable updates from server deployment to end-user integration.

CERTIFICATIONS

- **AWS Certified Cloud Practitioner** – Amazon Web Services (AWS) 2026

TECHNICAL SKILLS

- **Computer Science Fundamentals:** OOP, Data Structures, Algorithms, Design Patterns, SOLID Principles, Problem Solving
- **Programming Languages:** C/C++, Python, Java, SQL, Bash, HTML, CSS, JavaScript, TypeScript
- **Software Engineering:** Software Architecture, SDLC, Agile/Scrum, Technical Documentation, Git, GitHub, GitLab
- **Systems & Linux:** Linux Programming, Bash Scripting, CMake, IPC, Signal Handling, Daemon Services, Embedded Linux (Buildroot, Yocto)
- **Frontend Development:** Angular, Reactive Forms, RxJS, Bootstrap 5, Responsive Design, JWT Auth Flows, REST API Integration, Cross-Browser Debugging
- **Backend Development:** Node.js, Express.js, Spring Boot, MongoDB, Mongoose, REST APIs, JWT Authentication, Middleware Design, Joi Validation, Flask
- **Testing & QA:** ISTQB Foundation, Manual & Automated Testing, Selenium, TestNG, JUnit, Postman, API Testing, Allure Reports, Jira
- **DevOps & Cloud:** AWS Cloud Practitioner, Docker, Docker Compose, Jenkins, GitHub Actions, GitLab CI/CD, Terraform, Kubernetes, Ansible, ArgoCD, Grafana, Prometheus

INTERNSHIPS

EME (Egypt Makes Electronics) – Embedded Linux Internship

Jun 2024 – Mar 2025

Linux System Programming and Embedded Linux

Alexandria, Egypt

- Developed software using Python and Modern C++, applying OOP principles. Worked with Raspberry Pi for IoT and image processing with OpenCV.
- Gained experience in Linux system programming, including kernel customization using Buildroot and Yocto.
- Worked with TCP/IP, socket programming, POSIX threads, and Linux kernel modules.

- Utilized Bash scripting, Signal Handling, IPC, Daemon Services, and tools like CMake, Qt, Node-RED, and MQTT for embedded and networked systems.

Career180

Oct 2024 – Mar 2025

Software Testing Internship

Alexandria, Egypt

- ISTQB Foundation Level, Manual Testing, Agile Software Development, Database, Web and Mobile Application Automation, Automation Testing with Selenium.

DEPI (Digital Egypt Pioneers Initiative)

Apr 2024 – Oct 2024

Software Tester Internship

Alexandria, Egypt

- Gained in-depth knowledge and practical experience in both manual and automated software testing.
- Worked on real-world projects like Swag Labs, automationexercise site, and Go Bus, designing and executing test plans, automated test scripts, writing test cases, and reporting software defects.
- Applied Agile methodologies and followed the full software development life cycle (SDLC).

Siemens Classical AUTOSAR Fundamentals

Jul 2024 – Dec 2024

Onsite AUTOSAR Training

Alexandria, Egypt

- 3 months onsite training at Siemens, working with tools like Generators, Configurations, hardware projects, and debuggers like Lauterbach, providing both conceptual and hands-on experience.

Samsung Innovation Campus (IoT Training)

Jun 2024 – Oct 2024

Onsite IoT Training

Cairo, Egypt

- 3 months hybrid training (Online/Offline) on the foundations of IoT provided by Samsung Electronics, focused on developing IoT applications.
- Covered IoT Platforms, Raspberry Pi, Node.js, Linux System Programming, ThingSpeak, and MQTT.

EME – IoT Training

Aug 2023 – Nov 2023

IoT Training on ESP32 and Thinger.IO

Alexandria, Egypt

- WiFi Concept in ESP32, Thinger.IO IoT, MP/HTTP/MQTT, NTP, HTTP Request, Node-RED Platform, MQTT Web Server, Node-RED Dashboard with MQTT, ThingSpeak HTTP/MQTT, Postman.
- Introduction to IIoT, Modbus, and Modbus TCP/IP.

Siemens Embedded Systems with DevOps Summer Training

Jun 2022 – Aug 2022

Developing Embedded Software Application with DevOps Guide

Alexandria, Egypt

- MCU Fundamentals, RTOS Fundamentals, MISRA C Rules, Functional Safety (ISO 26262), RTE Generator, AUTOSAR Layers and Methodologies, DevOps, AUTOSAR CAN Stack, Bootloader, OSEK AUTOSAR OS Fundamentals.
- CI/CD: Worked with Jenkins, GitLab, and GitHub Actions to automate pipelines. IaC & Configuration Management: Learned Terraform and Ansible. Code Quality: Integrated tools like SonarQube and Coverity into CI/CD.

PROJECTS

Online Bookstore (MEAN Stack)

2025

Full-Stack E-Commerce Web Application

- Built the **backend authentication system** with **JWT access/refresh token rotation**, **bcrypt password hashing**, **Joi request validation**, and **role-based authorization** middleware (user/admin) using **Node.js**, **Express.js**, and **MongoDB/Mongoose**. Developed **RESTful user APIs** for profile management, password updates, and admin user operations with **pagination**.
- Implemented the **Angular 19 frontend auth layer: token service** (JWT decode, expiry checks, localStorage management), **HTTP interceptor** with **silent token refresh** and **concurrent request queuing**, **functional route guards** (auth/admin/guest), and reactive auth state via **BehaviorSubject**. Built **login**, **registration**, **email verification**, **profile management**, **password change**, and **order history** pages using **standalone components**, **reactive forms**, **lazy-loaded routes**, and **Bootstrap 5**.
- Responsible for **deployment on AWS**, including configuring the production environment, deploying backend services, and ensuring proper hosting and connectivity between the frontend, backend, and database.
- Configured and managed **MongoDB Atlas (cloud database)**, including cluster setup, network access control (IP whitelisting), database users and roles, connection string integration, and ensuring secure and scalable database connectivity for the application.

Bash Shell DBMS

2025

File-Based Database Management System in Bash

- GitHub Project : Built the **database management layer** (create, list, connect, drop databases) using **directories as storage** on the Linux filesystem. Developed the **CLI menu system**, shared **utility libraries**, **operation logging**, and **backup/restore** functionality. Designed a **modular architecture** enabling parallel development with **zero merge conflicts** using a branched **Git workflow**.

Cloud DevOps Pipeline Project

2025

End-to-End CI/CD with Hybrid Kubernetes and AWS

- GitHub Project : Designed and implemented a complete DevOps pipeline using **Terraform** for AWS infrastructure provisioning and **Ansible** for automated configuration. Integrated **Jenkins (CI)** with **Docker** and **Trivy** for image building and security scanning, while enabling GitOps-based **Continuous Deployment with ArgoCD (CD)**. Deployed workloads on a hybrid **Kubernetes (kubeadm)** cluster securely connected via **Tailscale VPN**. Configured monitoring and observability using **Prometheus** and **Grafana**, ensuring a production-grade automated pipeline.

Scalable E-Commerce Platform

2025

CI/CD Deployment and Monitoring of Microservices on AWS

- GitHub Project : Built and deployed a containerized e-commerce microservices application using Jenkins, Docker, and Maven with automated CI/CD workflows. Provisioned AWS infrastructure using Terraform and configured EC2 instances with Ansible for environment setup and deployment. Integrated Kubernetes for orchestration, Argo CD for GitOps-based continuous deployment, and Prometheus/Grafana for real-time monitoring and alerting. Managed source control via Git and GitHub, ensuring automated testing and delivery pipelines for high system availability.

AWS Infrastructure Automation with Terraform

2025

Milestone Project in ITI - DevOps Track

- GitHub Project : Implemented secure AWS VPC with public/private subnets, EC2 instances, and security groups using Terraform. Configured route tables for secure communication and internet access for public subnet instances. Automated resource management and deployment with Terraform, focusing on instance access control and network security.

WiFi - Bluetooth Controll

2024

Final Project in C++ Module Provided from EME Internship

- GitHub Project: Command-line tool designed to streamline WiFi and Bluetooth management on Linux systems in an easy and efficient way. This program provides complete control over WiFi - Bluetooth network from just the terminal.

Simple Linux Shell

2024

debut lab assignment in the operating systems course this semester

- GitHub Project: A lightweight, user-friendly UNIX shell built using C. Designed to streamline user interaction with the operating system through a sleek command-line interface. Features include built-in commands and seamless execution of external programs for enhanced productivity.

Languages

Arabic: Native Proficiency

English: Very Good Reading and Writing Proficiency